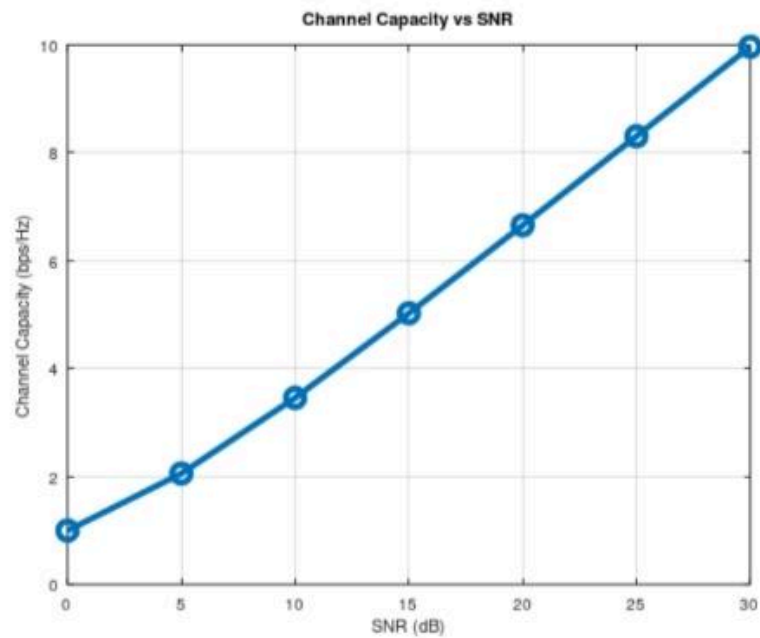


EX-9 Massive MIMO Capacity vs SNR and Gain Usi
Objective: Analyze Channel Capacity (bps/Hz)

CODE :

```
1 clc;
2 clear;
3 close all;
4 snr = 0.5:30;
5 snr_linear = 10.^(snr/10);
6 capacity = log2(1 + snr_linear);
7 figure;
8 plot(snr,capacity,'b','o','LineWidth',2,'MarkerSize',6);
9 grid on;
10 xlabel('SNR (dB)');
11 ylabel('Channel Capacity (bps/Hz)');
12 title('Channel Capacity vs SNR');
13
```

OUTPUT :

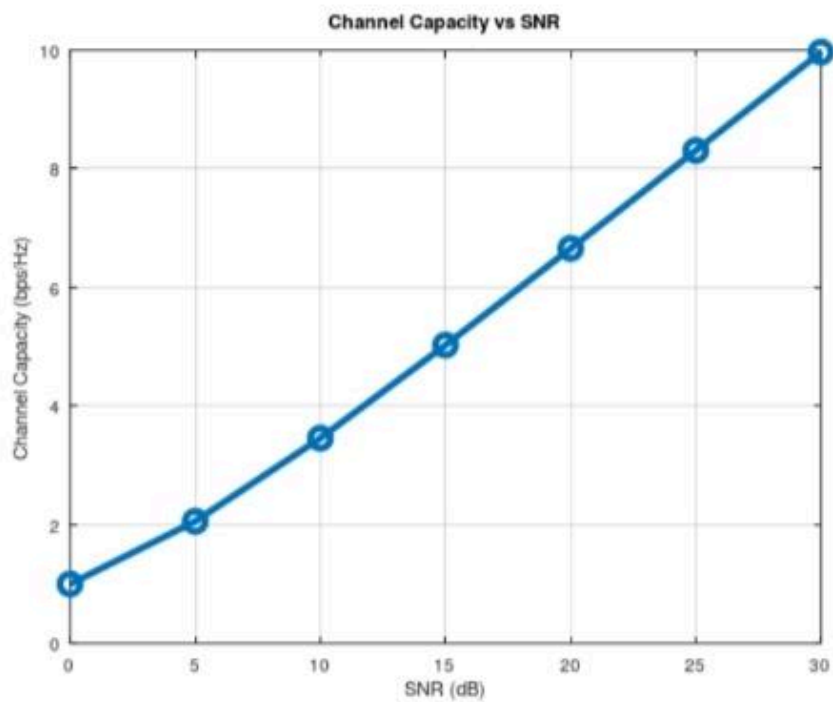


Objective Comparison of Additive White Gaussian Noise (AWGN) Channel Capacity vs SNR (dB)

C O D E :

```
1  clc;
2  clear;
3  close all;
4  snr = 0:5:30;
5  snr_linear = 10.^(snr/10);
6  capacity = log2(1 + snr_linear);
7  figure;
8  plot(snr, capacity, 'b', 'LineWidth', 2, 'MarkerSize', 6);
9  grid on;
10 xlabel('SNR (dB)');
11 ylabel('Channel Capacity (bps/Hz)');
12 title('Channel Capacity vs SNR');
13
```

O U T P U T :



Objective: Analyze Antenna Gain (dB)

CODE :

```
1  clc
2  clear
3  close all
4  antennas = [4 8 16 32 64]; % Number of Antenna Elements
5  gain = 10*log10(antennas); % Antenna Gain (dB)
6  figure
7  bar(antennas, gain)
8  grid on
9  xlabel('Number of Antenna Elements')
10 ylabel('Antenna Gain (dB)')
11 title('Antenna Gain for Different Antenna Array Sizes')
```

OUTPUT :

